

Homework — 1.5.2 Moral and Ethical Issues

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.5.2 lesson.

Part A — This week's topic (~14 marks)

1. A hospital introduces an AI system that reads scans and flags likely tumours for a radiologist to check.

(a) State **one** benefit of using automated decision making here. **[1]** (AO2)

(b) Explain **one** ethical concern with relying on this system, referring to a named issue from 1.5.2. **[1]** (AO2)

2. A facial-recognition system used by a police force is found to misidentify people from some ethnic groups far more often than others.

(a) Explain **why** an AI system can end up biased in this way. **[2]** (AO2)

(b) Explain why the "black box" nature of such a system makes the problem harder to deal with. **[1]** (AO1/AO2)

3. A tech company moves its software-development team from the UK to a country with lower wages (offshoring). State **one** benefit to the company and **one** drawback that this could cause. **[2]** (AO2)

4. A streaming firm replaces its proprietary media player with components built on **open-source** software. Explain **one** way open-source software can counter the problems caused by a single company holding a **monopoly**. [2] (AO2)

5. A council moves all of its services — benefits claims, bin collections, parking permits — to "online only". **Discuss** the moral and social impact of this decision on the different groups affected, and reach a justified conclusion. [5] (AO3)

Part B — Retrieval: keep it fresh (~6 marks)

6. (1.5.1) A hacker breaks into a company server *intending* to steal trade secrets to sell to a rival. State the Act and the specific **offence** committed. [2] (AO1)

7. (1.3.3) State the purpose of the **TCP** layer and the purpose of the **IP** layer in the TCP/IP stack. [2]
(AO1)

8. (1.1.2) State **one** reason a **multi-core** processor can complete a workload faster than a single-core processor of the same clock speed. [2] (AO1)
